PRODUCT DATA SHEET

nXQ55i DRY MULTISTAGE ROOTS PUMP

edwardsvacuum.com

The nXQ55i is designed for environments where gas containment is essential, providing a remarkably low leak rate < 5×10^{-8} mbar·l/s.
Its Multistage Roots design provides a dry, dust-free environment, enabling clean operation for sensitive applications.
The nXQ55i is optimised to support gas retention and minimise contamination, which is particularly useful in helium recirculation systems.



With a focus on reliability and system integrity, it provides a robust solution for advanced applications such as:

- Helium-4 cryostats
- Dilution refrigeration systems
- Leak detection setups

Leak-tight design
System integrity!

Strong light gas performance
Ideal for precision applications!

Dry pump design
For clean processes!

FEATURES AND BENEFITS

1 The nXQ55i is developed to satisfy leak-tightness standards, achieving rates below 5×10^{-8} mbar·l/s. **This level helps minimise gas loss ensuring consistent performance.**

As part of a rigorous quality process, each nXQ55i is subjected to an integral leak test, and every pump is supplied with a certificate confirming the test has been successfully completed and validated. This guarantees reliability and customer confidence.

2 Engineered for rare gases such as helium-3 and helium-4, the nXQ55i delivers high pumping speed (up to 52 m³/h for helium) supporting efficient recovery and recirculation in demanding environments. **This ensures stable operation even under continuous use**, making it suitable for applications where gas purity and conservation are important.

3 Its compact dry pump design is dust-free, ensuring clean processes and results while avoiding unpleasant odours and particulates for a more comfortable working environment.

Low noise levels, reduced energy consumption, and the absence of oil changes or disposal requirements make it a cleaner alternative to traditional “wet” pumps.

With low maintenance requirements for up to five years, the nXQ55i contributes to reduced downtime and, ultimately, a low cost of ownership.

TECHNICAL DATA

	nXQ55i
Maximum nitrogen pumping speed	55 m^3/h
Maximum helium pumping speed	52 m^3/h
Ultimate pressure	< 0.03 mbar
Inlet flange ISO-KF	NW40
Outlet flange ISO-KF	NW16
Maximum permitted outlet pressure	200 mbarg
Maximum continuous inlet pressure	1050 mbar*
Operating temperature range	+5 to +40°C
Nominal rotational speed	12600 rpm
Voltage input	100-120 V
	200-240 V
Mains frequency	50/60 Hz
Power connector 1-ph	C 20 acc. to IEC 60320
Power at ultimate	600 W
Leak tightness	< 5 x 10^{-8} mbar·l/s
Weight, approx.	43 kg
Noise at ultimate vacuum	< 52 dB(A)

* For installation heights above 2000 m, an inlet pressure of 50 mbar should not be permanently exceeded.

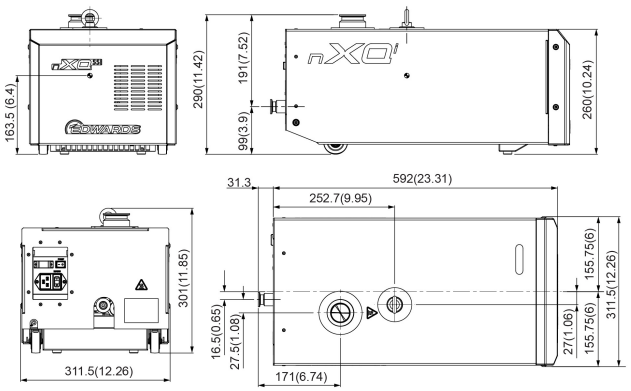
ORDERING INFORMATION

Product Description	Order number
nXQ55i - 100-120V / 1-ph / 50/60Hz	A78051310
nXQ55i - 200-240V / 1-ph / 50/60Hz	A78052310

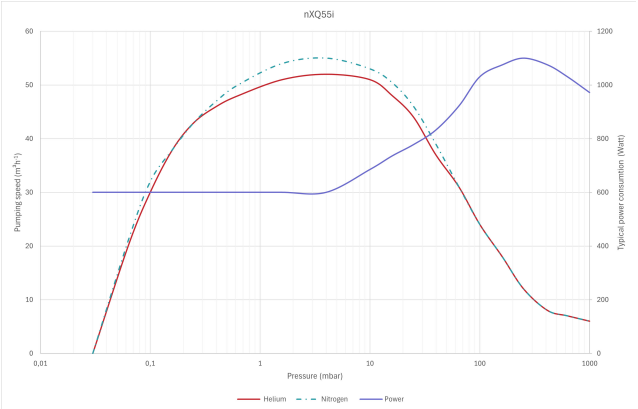
ACCESSORIES

Product Description	Order number
2m electrical supply cable, NEMA 5-15P, North America/Japan (100-120 V)	A50507003
2m electrical supply cable, NEMA 6-15P, North America (200-240 V)	A50507006
2m electrical supply cable, UK, three pin plug	A50505003
2m electrical supply cable, Europe	A50506003
2m electrical supply cable, China	A50509003
USB Interface Cable	A90590005
Replacement interface bridging plug	A90590006
Inlet screen DN 40 ISO-KF	C10516085

DIMENSIONS



PUMPING SPEED DEPENDENCE ON PRESSURE



Publication Number: 3601 0855 01
© Edwards Limited 2026. All rights reserved.
Edwards and the Edwards logo are trademarks of Edwards Limited.

Whilst we make every effort to ensure that we accurately describe our products and services, we give no guarantee as to the accuracy or completeness of any information provided in this datasheet.

Edwards Ltd, registered in England and Wales No. 6124750, registered office: Innovation Drive, Burgess Hill, West Sussex, RH15 9TW, UK.